

Practice Guidance



Semaglutide therapy for metabolic dysfunction–associated steatohepatitis: November 2025 updates to AASLD Practice Guidance

Abstract

This practice recommendation serves as an update to the 2023 AASLD Practice Guidance on the clinical assessment and management of nonalcoholic fatty liver disease (NAFLD), now known as metabolic dysfunction–associated steatotic liver disease (MASLD), and provides implementable guidance on patient selection for treatment, consideration of comorbidities, and monitoring of treatment safety and efficacy of semaglutide. *FDA-indication and Practice Recommendation:* The Wegovy formulation, whose main ingredient is semaglutide, received accelerated FDA approval in August 2025 for treating MASH with moderate-to-advanced fibrosis (consistent with stages F2–F3 fibrosis), based on interim results of the phase 3 ESSENCE trial where 72 weeks of 2.4 mg/week subcutaneous injection resulted in achievement of both primary histologic endpoints: (1) resolution of MASH without worsening of fibrosis (62.9% vs. 34.3% placebo, $p < 0.001$) and (2) ≥ 1 stage reduction in liver fibrosis without worsening of MASH (36.8% vs. 22.4% placebo, $p < 0.001$); final approval awaits long-term outcomes. *Patient Selection:* Candidates should have MASH with stage 2–3 fibrosis, identified using non-invasive tests (NITs) such as VCTE (8–15 kPa), MRE (3.1–4.4 kPa), or ELF (9.2–10.5), rather than liver biopsy, which is impractical and unnecessary for most patients. In those with VCTE (15–20 kPa), MRE (4.4–5 kPa), or ELF (10.5–11.3), an individualized decision to treat should be based on exclusion of cirrhosis with another confirmatory NIT, or cross-sectional imaging excluding nodular-appearing liver contour and signs of portal hypertension, or a

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Abbreviations: AASLD, American Association for the Study of Liver Diseases; ADA, American Diabetes Association; AgRP, agouti-related peptide; ALT, alanine aminotransferase; AMD, age-related macular degeneration; AST, aspartate transaminase; AUD, alcohol use disorder; AUDIT-C, Alcohol Use Disorders Identification Test–Consumption; CAP, controlled attenuation parameter; CCK, cholecystokinin; CV, cardiovascular; CVD, cardiovascular disease; cT1, iron-corrected T1 mapping; DEXA, dual-energy x-ray absorptiometry; DPP-4, dipeptidyl peptidase-4; ELF, enhanced liver fibrosis test; FDA, U.S. Food and Drug Administration; FIB-4, fibrosis-4 index; FLOW, Evaluate Renal Function with Semaglutide Once Weekly; GI, gastrointestinal; GIP, glucose-dependent insulintropic polypeptide; GLP-1R, glucagon-like peptide-1 receptor; GLP-1 RAs, glucagon-like peptide-1 receptor agonists; kPa, kilopascals; LSM, liver stiffness measurement; MASH, metabolic dysfunction–associated steatohepatitis; MASLD, metabolic dysfunction–associated steatotic liver disease; MEN2, multiple endocrine neoplasia type 2; MetALD, metabolic dysfunction–associated alcohol-related liver disease; MRI, magnetic resonance imaging; MRE, magnetic resonance elastography; MTC, medullary thyroid carcinoma; NAFLD, non-alcoholic fatty liver disease; NAION, non-arteritic anterior ischemic optic neuropathy; NASH, non-alcoholic steatohepatitis; NILDA, non-invasive liver disease assessments; NITs, non-invasive tests; OSA, obstructive sleep apnea; PCP, primary care provider; PDFF, proton density fat fraction; PETH, phosphatidylethanol; PP, pancreatic polypeptide; PYY, peptide YY; RCT, randomized controlled trial; SELECT, Semaglutide Effects on Cardiovascular Outcomes in People with Overweight or Obesity; SGLT2, sodium-glucose co-transporter-2; STEP, Semaglutide Treatment Effect in People with Obesity; SUSTAIN-6, Trial to Evaluate Cardiovascular and Other Long-term Outcomes with Semaglutide in Subjects with Type 2 Diabetes; SWE, shear wave elastography; T2DM, type 2 diabetes mellitus; VCTE, vibration-controlled transient elastography; VLDL, very-low-density lipoprotein.

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platelet count of $< 150,000/\text{mm}^3$. While semaglutide is not approved to treat patients with MASH cirrhosis (VCTE > 20 kPa, MRE > 5.0 kPa, ELF > 11.3 , and/or evidence of portal hypertension), those with compensated cirrhosis who are receiving semaglutide for another FDA-approved indication should be monitored carefully. *Monitoring and Safety:* Semaglutide showed a favorable hepatic safety profile in the ESSENCE trial, with no discontinuations due to liver enzyme elevations. Routine hepatic panels are recommended only as clinically indicated. The most common adverse events were gastrointestinal (nausea, diarrhea, constipation, vomiting), generally mild and transient; patient education and dose titration help improve tolerance. Clinicians should monitor for rare but serious risks, including acute kidney injury (from dehydration), symptomatic gallbladder disease, pancreatitis, thyroid C-cell tumors, retinopathy progression, and lean mass loss. *Treatment Response and Concomitant Therapy:* Lifestyle modification remains the cornerstone of MASLD/MASH management alongside semaglutide. Combination use of resmetirom with semaglutide 2.4mg/week has not been studied. While no NITs reliably predict histologic response at the individual patient level, reductions from baseline to 72 weeks of treatment suggest significant improvement in MASH resolution (ALT ≥ 17 U/L or $\geq 20\%$) and fibrosis improvement (VCTE LSM $\geq 30\%$; MRE LSM $\geq 20\%$; ELF ≥ 0.5). Non-response may be suspected if ALT or NITs worsen. The benefit is uncertain if a suboptimal response occurs, and may require an individualized approach and further follow-up.

Keywords: fatty liver disease, MASH, MASLD, NAFLD, NASH, semaglutide, Wegovy

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BACKGROUND

This practice recommendation serves as an update to the 2023 AASLD Practice Guidance on the clinical assessment and management of nonalcoholic fatty liver disease, now known as metabolic dysfunction–associated steatotic liver disease (MASLD),^[1] and provides implementable guidance on patient selection for treatment, consideration of comorbidities, and monitoring of treatment safety and efficacy of semaglutide. The Wegovy formulation, whose main ingredient is semaglutide, received accelerated approval from the United States Food and Drug Administration (FDA) on August 15, 2025, to treat metabolic dysfunction–associated steatohepatitis (MASH), formerly known as nonalcoholic steatohepatitis (NASH), in adults with moderate-to-advanced fibrosis (FDA label^[2]). Semaglutide (Wegovy) was initially approved in 2021 for chronic weight management in adults with obesity or overweight accompanied by at least 1 weight-related comorbidity; this indication was extended in December

2022 to include pediatric patients aged 12 years and older. In March 2024, semaglutide (Wegovy) was approved to reduce the risk of major adverse cardiovascular (CV) events (CV death, nonfatal myocardial infarction, or nonfatal stroke) in adults with established CV disease and either obesity or overweight. The approval to treat MASH in adults with moderate-to-advanced fibrosis (consistent with stages F2–F3) was based on the findings from the first 800 participants enrolled in the Phase 3 ESSENCE trial (clinicaltrials.gov/study/NCT04822181) in which 72 weeks of semaglutide 2.4 mg/week subcutaneous injection achieved both primary histologic endpoints: (1) resolution of MASH without worsening of fibrosis (62.9% vs. 34.3% placebo, $p < 0.001$) and (2) ≥ 1 stage reduction in liver fibrosis without worsening of MASH (36.8% vs. 22.4% placebo, $p < 0.001$).^[3] Final approval to treat MASH with moderate-to-advanced fibrosis is contingent upon the verification and description of clinical benefit in the ongoing phase 3 ESSENCE clinical trial, results of which are anticipated in 2028 or thereafter.

The writing group, commissioned by the AASLD (American Association for the Study of Liver Diseases) governing board, examined data from the ESSENCE trial on invasive (liver biopsy) and non-invasive tests (NITs) for liver fibrosis concerning both patient selection and treatment response. We also reviewed published data on semaglutide for MASH and other FDA-approved indications, along with publicly available FDA documents and selected unpublished data provided by Novo Nordisk (2025). While the FDA prescribing information is the official regulatory document for use in the United States, this guidance serves as a practical supplement for clinicians to help guide the management of patients with MASH who are eligible for semaglutide (Wegovy).

WHOM TO TREAT

The ESSENCE trial enrolled patients with biopsy-confirmed MASH and Metavir stages 2–3 fibrosis (F2–F3). There was no prerequisite number of metabolic comorbidities or specific NIT cutoffs for trial screening, and mild alcohol use was allowed (up to 20/30 g/d in women/men, consistent with most MASH trials criteria). Because liver biopsy will not be feasible or practical as a basis for patient selection in most patients who may be candidates for semaglutide, this guidance incorporates NIT-based criteria to identify appropriate candidates with MASH and stages 2–3 fibrosis in clinical practice (Figure 1).

It is important to note that all the elements that define fibrotic MASH, as specified in clinical trials—steatosis, ballooning, inflammation, and fibrosis—cannot be determined definitively without a liver biopsy. However, steatosis can be predicted with varying degrees of accuracy using imaging-based NITs, whereas fibrosis (with or without steatohepatitis) can be estimated using imaging and/or blood-based NITs with a reasonable-to-high accuracy, and are already widely used in clinical practice. Furthermore, the FDA approval of semaglutide does not include a requirement for a liver biopsy to define fibrotic MASH. Thus, using NIT data available from ESSENCE and extrapolating from other studies for those NITs not evaluated in the semaglutide MASH trial, several NIT types and cutoff values can be recommended to select treatment candidates in clinical practice. For patients who have undergone a liver biopsy in the prior 6–12 months, the liver disease stage can be determined based on those histology findings, provided that no substantial weight change has occurred in the interim; however, obtaining a baseline NIT before initiating semaglutide would be beneficial to enable longitudinal non-invasive assessment of treatment response.

AASLD has published 2 relevant guidelines for the use of NITs to assess liver disease: one focusing on

blood-based tests and one focusing on imaging-based tests; thus, a detailed review of types and performance features of NITs is beyond the scope of this guidance.^[4,5] The initial step to rule out advanced fibrotic MASH when evaluating patients with MASLD in primary care and non-hepatology referral settings is the fibrosis-4 (FIB-4) test. A FIB-4 score below 1.3 excludes advanced fibrosis (F3 or greater) in 99% of all patients and 95% of patients with multiple cardio-metabolic risks in the primary care setting.^[6,7] In contrast, a FIB-4 ≥ 1.3 has a weaker positive predictive value for significant (F2 or greater) or advanced fibrosis. To be clear, FIB-4 should not be used as a single rule-in or rule-out test for semaglutide eligibility for patients with MASLD/MASH. Consequently, AASLD promotes a sequential testing strategy with elastography or blood-based NITs after calculating FIB-4, to evaluate liver disease stage and semaglutide treatment candidacy.^[1]

In ESSENCE, liver stiffness measurement (LSM) by vibration-controlled transient elastography (VCTE) was the only imaging-based NIT evaluated. VCTE can be performed as a point-of-care test and is widely available and accessible. Limitations include less reliable estimation of LSM in patients with severe obesity, especially BMI >40 kg/m², severe steatosis, and those with narrow rib spaces.^[8] Most studies support a cutoff of 8 kPa for detecting stage ≥ 2 fibrosis in MASH, a threshold also recommended as second-line screening in patients with FIB-4 > 1.3 ^[1] and for candidate selection in resmetirom therapy under AASLD guidance.^[9] At trial randomization, ESSENCE participants had a mean (SD) LSM of 12 (± 6) kPa (range 3.9–69.8). The median and interquartile range (IQR) of LSM by VCTE were 10.8 (8.5, 15.1) kPa in those assigned to semaglutide, and 11.1 (8.6, 16.1) kPa in those assigned to placebo (see Table 1 for distribution of LSM measurements at baseline and week 72 by treatment group).

This guidance supports the thresholds of LSM 8–15 kPa by VCTE for the selection of treatment candidates with semaglutide, as this is in line with the IQR of LSM by VCTE in ESSENCE and aligns with prior AASLD guidance on selecting individuals who meet the resmetirom treatment indications based on fibrosis stage. The recommended range of 8–15 kPa included ~50% of the study population, while up to 19% of participants had LSM < 8 kPa, and up to 32.5% of participants had LSM ≥ 15 kPa. Thus, to determine treatment eligibility of individuals with MASLD outside of 8–15 kPa, clinicians may use concordant values of additional available NITs. For those with LSM by VCTE of 15–20 kPa (present in up to 19% of ESSENCE participants), individualized decision to treat should be based on exclusion of cirrhosis with another confirmatory NIT, or cross-sectional imaging excluding nodular-appearing liver contour and signs of portal

Use of NITs for determining eligibility for semaglutide (Wegovy®) for the treatment of MASH with moderate-to-advanced fibrosis (F2-F3)

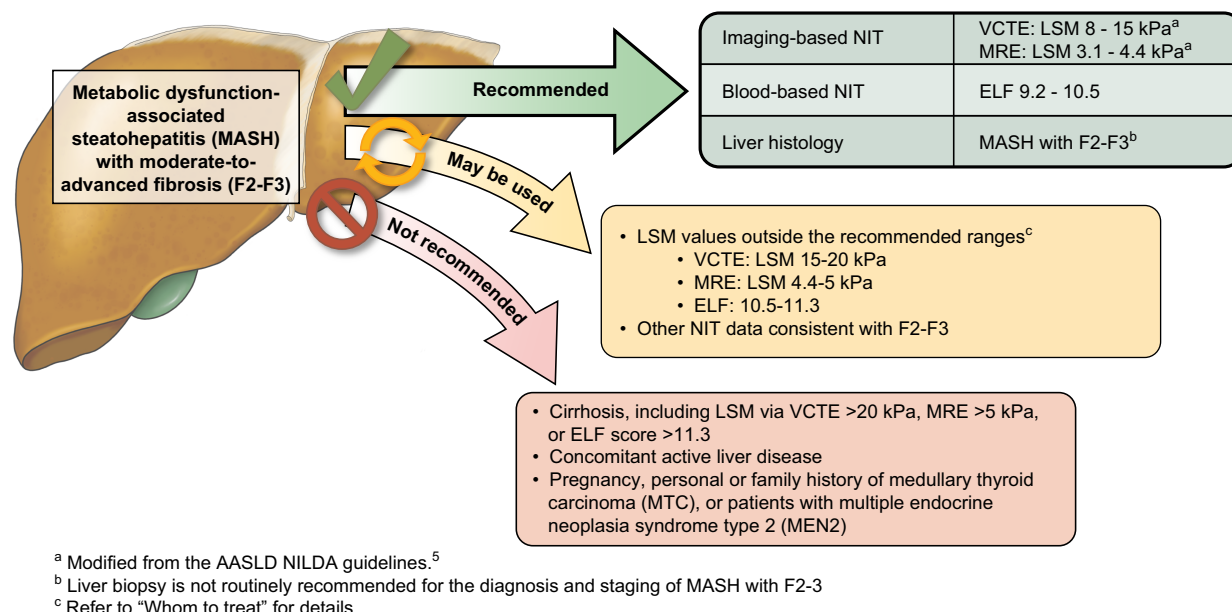


FIGURE 1 Use of NITs for determining eligibility for semaglutide (Wegovy) for the treatment of MASH with moderate-to-advanced fibrosis (F2-F3). Abbreviations: AASLD, American Association for the Study of Liver Diseases; ELF, enhanced liver fibrosis testing; LSM, liver stiffness measurement; MASH, metabolic dysfunction-associated steatohepatitis; MRE, magnetic resonance elastography; NILDA, non-invasive liver disease assessments; NITs, non-invasive tests; VCTE, vibration-controlled transient elastography.

hypertension, or a platelet count of $< 150,000/\text{mm}^3$.^[10] Semaglutide (Wegovy) has not been approved for the treatment of MASH cirrhosis and, thus, should not be prescribed solely for this indication. For those who qualify for semaglutide for other approved indications, data from a single phase 2 trial of semaglutide in patients with compensated cirrhosis did not identify liver-related safety concerns.^[8]

The most accurate assessment of liver fibrosis and steatosis is achieved with MRI-based techniques, which provide liver stiffness [magnetic resonance elastography (MRE)] and fat quantification [magnetic resonance imaging-proton density fat fraction (MRI-PDFF)]. Although MRE is less widely available than VCTE and requires a radiology suite, which is more costly, there are now ~1400 available in the United States. The short, non-contrast MRE provides the LSM, and the MRI-PDFF may be requested as part of it, and often avoids the cost and resources of a full MRI. MRE offers higher accuracy overall, especially in patients with central adiposity (BMI $> 35 \text{ kg/m}^2$), when VCTE results conflict with clinical impression, or when VCTE cannot be performed due to a narrow intercostal space. While MRE was not used in ESSENCE, MRE-based criteria may be extrapolated from prior studies and previous AASLD guidance^[5,11] with a recommended

range of 3.1–4.4 kPa to identify patients with moderate-to-advanced fibrosis consistent with stages 2–3 fibrosis. For those with LSM 4.5–5.0 kPa, exclusion of cirrhosis and portal hypertension should be considered, as described in the VCTE section.

Blood-based biomarkers such as the Enhanced Liver Fibrosis (ELF) score can also be used for the identification of candidates for semaglutide treatment.^[12] In ESSENCE, the mean (SD) ELF score was 9.95 (± 0.98), ranging from 7.4 to 13.2, similar to results from the MAESTRO-NASH trial of resmetirom. The median (IQR) ELF value was 9.9 (9.2, 10.5) in both semaglutide and placebo participants. Based on prior studies, a cutoff of 9.2 provides optimal sensitivity and specificity for detecting stage ≥ 2 fibrosis.^[13] In alignment with the IQR of ELF in ESSENCE, this guidance supports the range of 9.2–10.5 to guide the selection of candidates for semaglutide therapy. This range identified approximately 50% of participants in the ESSENCE trial with histological F2–3. Of note, among those with histological F2–3, 21% had ELF values < 9.2 , 15% had values 10.5–11.3, and 9% had > 11.3 . Therefore, these ranges serve as guidance. For those with ELF values 10.5–11.3, exclusion of cirrhosis and portal hypertension should be considered as described in the VCTE section.

TABLE 1 Aggregate results of selected non-invasive tests (NITs) according to treatment arm and, where available, histological response from the ESSENCE trial

NIT			Semaglutide 2.4mg (n=534)	Placebo (n=266)
ALT (U/L)	Week 72 relative change from baseline (%)		-52.10	-22.20
	Median change (IQR) from baseline to week 72	MASH resolution	-30.00 (-56.75; -12.25)	-10.00 (-1.00; 40.00)
		No MASH resolution	-23.00 (-48.50; -8.50)	-4.00 (-20.00; 9.00)
CAP (dB/m)	Week 72 absolute reduction from baseline		-43.40	-13.10
	Median change (IQR) from baseline to week 72	MASH resolution	-53.00 (-88.25; - 10.25)	-16.00 (-42.00; 21.00)
		No MASH resolution	-27.00 (-65.25; 3.25)	-16.00 (-43.50; 14.75)
LSM (kPa)	Week 72 relative change from baseline (%)		52.00%	30.30%
	Median change (IQR) from baseline to week 72	Improvement in liver fibrosis	-3.20 (-6.20; -1.05)	-2.40 (-4.90; -0.60)
		No improvement in fibrosis	-3.60 (-6.70; -1.20)	-1.30 (-3.60; 2.25)
ELF Score	Week 72 absolute change from baseline		-0.60	0.00
	Median change (IQR) from baseline to week 72	Improvement in liver fibrosis	-0.70 (-1.11; -0.12)	-0.24 (-0.67; 0.30)
		No improvement in fibrosis	-0.68 (-1.10; -0.10)	0.10 (-0.40; 0.50)
Proportion achieving ≥30% decrease in LSM at week 72			52.00%	30.30%
Proportion achieving ≥0.5 decrease in ELF score at week 72			55.80%	25.50%

Note: Improvement in liver fibrosis = improvement in liver fibrosis and no worsening of steatohepatitis; MASH resolution = resolution of steatohepatitis without worsening of fibrosis.

Abbreviations: ALT, alanine transaminase; CAP, controlled attenuation parameter; ELF, enhanced liver fibrosis; IQR, Interquartile range; LSM, Liver stiffness measurement; MASH, metabolic-associated steatohepatitis.

As the NIT research expands, additional biomarkers with demonstrated and validated high performance in detecting F2–3 should be considered for treatment selection and monitoring, adapted to local and regional availability. While a number of other NITs have shown promise in research settings, a comprehensive review of these additional tools is beyond the scope of this document and has been addressed in the AASLD NILDA (non-invasive liver disease assessment) guidance.^[4,5]

While patients with minimal alcohol use (up to 20/30 g/d in women/men) were enrolled in the ESSENCE trial, those with higher alcohol consumption [ie, in the MetALD (metabolic dysfunction and alcohol-associated liver disease) range] were excluded. Alcohol consumption should be assessed with both AUDIT-C (alcohol use disorders identification test-consumption) and Peth (phosphatidylethanol) testing.^[14] Given emerging data

on the role of glucagon-like peptide-1 receptor agonists (GLP-1 RAs) in decreasing alcohol cravings and as a potential treatment of alcohol use disorder (AUD),^[15,16] decisions to initiate semaglutide for the treatment of non-cirrhotic MASH in those with moderate alcohol consumption must be individualized. Where possible, patients should be enrolled in clinical trials.^[14] Importantly, NIT cut points may also differ in this population and thus must be considered in any decision to initiate treatment until additional data are available.

MONITORING DURING TREATMENT

Because the adverse events observed in the ESSENCE trial are consistent with those seen in trials for other approved Wegovy indications, the FDA recommends similar monitoring parameters.^[2] Since

ESSENCE was designed to evaluate semaglutide in participants with underlying liver disease, hepatic function tests were measured every 12 weeks, and criteria were established for discontinuation in participants with an increase in pre-specified changes in ALT, AST, total bilirubin, or INR (Supplemental Table S1, <http://links.lww.com/HEP/K318>). Among the 538 participants receiving semaglutide, none stopped it because of elevated ALT or AST. While ESSENCE monitored hepatic function tests every 12 weeks, semaglutide has demonstrated a favorable hepatic safety profile in this and prior studies. Based on these data, we recommend a hepatic function panel as clinically indicated rather than as a strict monitoring plan among patients receiving semaglutide for treatment of MASH.

Patients should be monitored for adverse events commonly associated with semaglutide use (Table 2). For those with active suicidal ideation, semaglutide should not be initiated until the patient is stable and under the care of a mental health professional. The adverse events that were most common among participants receiving semaglutide in the ESSENCE trial were nausea, diarrhea, constipation, and vomiting (see “Adverse events and safety monitoring” section; Supplemental Table S2, <http://links.lww.com/HEP/K319>), all of which were reported in prior trials of semaglutide.

Adverse events and safety monitoring

Overall, 86.2% of patients treated with semaglutide experienced at least 1 adverse event compared with 79.7% in the placebo group in the ESSENCE trial. Despite the higher overall incidence of adverse events, discontinuation rates due to adverse events were similar between the semaglutide and placebo groups (2.6% vs. 3.3%). In patients with MASH, semaglutide was well tolerated, with no increase in serious adverse events or treatment discontinuations compared with placebo, and similar mortality rates, supporting its favorable safety profile.^[3,17,18]

In the ESSENCE trial, the most common treatment-emergent adverse events were gastrointestinal (GI) disorders (Supplemental Table S2, <http://links.lww.com/HEP/K319>). Compared with placebo, semaglutide significantly increased incidences of nausea (36.2% vs. 13.2%), diarrhea (26.9% vs. 12.2%), constipation (22.2% vs. 8.4%), and vomiting (18.6% vs. 5.6%), as well as decreased appetite (14.0% vs. 2.8%). Most GI side effects were transient and tended to resolve over time, with most cases being mild or moderate in severity. However, on occasion, severe GI adverse reactions have occurred, leading to permanent discontinuation^[3] (Wegovy Prescribing Information^[2]).

Comprehensive patient education is vital to ensure adherence and reduce GI adverse effects. Patients

TABLE 2 Safety and efficacy assessments at baseline and on treatment with semaglutide

Timeframe	Safety assessments		Efficacy assessments		
	Symptoms	Diagnostic tests	Hepatic function panel	Non-invasive measurement of fibrosis	Non-invasive measurement of steatosis
Before treatment initiation	Active suicidal thoughts	- Evaluation for thyroid nodules - Consider retinal exam prior to GLP-1 RA therapy in patients with T2DM if not performed in past 12 months	X	X	X
On treatment monitoring	Nausea, vomiting, diarrhea, constipation, abdominal pain, depression or suicidal thoughts	As clinically indicated: - Pregnancy test - Hepatic function panel - RUQUS for symptomatic gallstone disease			
72 Weeks		Retinal exam as per society guidelines	X	X	X

Abbreviations: GLP-1 RA, glucagon-like peptide-1 receptor agonists; RUQUS, right upper quadrant ultrasound; T2DM, type 2 diabetes mellitus.

should be informed that semaglutide increases early satiety, leading to lower caloric intake, and should stop eating when feeling full to avoid nausea, bloating, or vomiting. Management strategies include gradual dose titration, smaller, lower-fat meals, adequate hydration, increased physical activity, and dietary fiber to ease constipation. Persistent or severe GI symptoms may require temporary dose reduction, slower titration, or supportive medications such as antiemetics, prokinetics, or antidiarrheals ([19], Wegovy Prescribing Information[2]).

There are a number of additional safety considerations for patients treated with semaglutide, which are detailed in the package insert (Wegovy Prescribing Information[2]). Here, we briefly highlight aspects for clinicians to be aware of when prescribing semaglutide (Wegovy) and outline a monitoring strategy in Table 3.

Acute kidney injury due to volume depletion can be seen secondary to GI adverse events. These GI effects are most prevalent when first starting the medication or when increasing the dosage. In patients with persistent nausea, vomiting, or diarrhea on GLP-1 RAs, periodic measurement of serum creatinine and estimated glomerular filtration rate helps detect early volume depletion and related renal impairment (Wegovy Prescribing Information[2]).

Cases of acute pancreatitis, including rare cases of necrotizing pancreatitis, have been reported in patients treated with semaglutide and other GLP-1 RAs, with several large meta-analyses showing conflicting results regarding the association, and recent large meta-analyses generally show no increased risk.[20] However, case reports describe pancreatitis occurring within 30 days to 5 weeks of GLP-1 RA initiation. Current prescribing guidelines recommend discontinuing semaglutide promptly if pancreatitis is suspected and avoiding re-initiation after a confirmed episode, regardless of cause. Current FDA prescribing information and clinical guidelines state that semaglutide should not be prescribed to patients with a history of pancreatitis[19] (Wegovy Prescribing Information[2]).

GLP-1 RAs are associated with a significantly increased risk of gallbladder disease. Cholelithiasis was reported by 1.6% of semaglutide-treated patients and 0.7% of placebo-treated patients. Cholecystitis was reported by 0.6% of semaglutide-treated patients and 0.2% of placebo-treated patients (Wegovy Prescribing Information[2]). A comprehensive meta-analysis of 76 randomized controlled trials involving 103,371 patients demonstrated a 37% increased risk of gallbladder or biliary diseases, with increases in cholelithiasis, cholecystitis, and cholecystectomy rates. The risk was dose-dependent and duration-dependent, and higher in weight loss trials compared with diabetes treatment trials.[21] A history of gallbladder disease is not an absolute contraindication; however, caution is

warranted, and patients should be educated on symptoms concerning for biliary complications[19] (Wegovy Prescribing Information[2]).

Semaglutide, like other GLP-1 RAs, carries a black box warning for the risk of thyroid C-cell tumors, including medullary thyroid carcinoma (MTC). Semaglutide is contraindicated in individuals with a personal or family history of MTC or multiple endocrine neoplasia syndrome type 2 (MEN2). In rodent studies, exposure to clinically relevant doses of semaglutide resulted in an increase in C-cell tumors. However, the relevance of these findings to humans remains uncertain. Rodents have a higher density of thyroid C cells and greater glucagon-like peptide-1 receptor (GLP-1R) expression, leading to increased calcitonin secretion and C-cell hyperplasia. In contrast, non-human primates and humans have fewer C cells and less GLP-1R expression, resulting in negligible stimulation of calcitonin release, even at supratherapeutic doses. Semaglutide does not have a meaningful effect on serum calcitonin levels in humans. At present, there is no conclusive evidence that semaglutide causes MTC in humans[22] (Wegovy Prescribing Information[2]). Pharmacovigilance data from the European Medicines Agency found no conclusive link between GLP-1 RAs and other types of thyroid cancer.[23]

Hypoglycemia risk is generally low with semaglutide monotherapy due to its glucose-dependent mechanism of action. However, the risk of hypoglycemia increases when used concomitantly with insulin or sulfonylureas. In the ESSENCE trial, hypoglycemia was reported in 7.4% of patients with type 2 diabetes mellitus (T2DM) receiving semaglutide versus 5.4% receiving placebo. Hypoglycemia has also been reported in patients without diabetes using GLP-1 RAs for weight management, particularly those with a history of bariatric surgery[3] (Wegovy Prescribing Information[2]).

Semaglutide has direct chronotropic effects on the heart through GLP-1R located in pacemaker cells of the sinoatrial node and is associated with increases in resting heart rate. Patients should be instructed to report palpitations or feelings of a racing heartbeat while at rest.[24]

Semaglutide induces a reversible delay in gastric emptying and, in rare cases, can precipitate gastroparesis with symptoms of nausea, vomiting, early satiety, and retained gastric contents. A 2025 retrospective analysis reported an incidence of 6.5 events per 1000 person-years among semaglutide users.[25] Accordingly, the use of semaglutide in severe gastroparesis is contraindicated, and expert panels advise baseline gastric motility assessment in patients with pre-existing motility disorders (Wegovy Prescribing Information[2]).

Delayed gastric emptying with GLP-1 RAs poses procedural hazards, and there is a theoretical concern for increased risk of aspiration in anesthetized patients,

TABLE 3 Key safety considerations and monitoring for semaglutide (Wegovy)

Safety concern	Monitoring/risk factors	Supporting data (ref#)	Clinical actions/recommendations
Acute kidney injury (AKI)	Check serum creatinine and eGFR at baseline, during initiation and titration. Higher risk if vomiting, diarrhea, or dehydration.	Volume depletion from GI effects noted in trials and postmarketing cases [Wegovy PI ⁽²⁾].	Ensure adequate hydration. Hold or reduce dose during significant GI intolerance. Reassess renal function until recovery.
Acute pancreatitis	Monitor for severe, persistent abdominal pain (± vomiting). Check amylase/lipase if symptomatic.	Case reports: onset within 30 days-5 weeks of initiation. Meta-analyses show no consistent increased risk ^(19,20) .	Discontinue immediately if suspected. Avoid rechallenge after confirmed episode.
Gallbladder disease (cholelithiasis/cholecystitis)	Monitor for right upper quadrant pain or biliary colic. Ultrasound if symptomatic.	Incidence: 1.60% (semaglutide) vs 0.70% (placebo). Meta-analysis (76 RCTs, n=103,371) showed 37% increased risk ^(19,21) .	Educate patients on biliary symptoms. Use caution in patients with history of gallbladder disease.
Thyroid C-cell tumors (MTC risk)	Review history of MTC or MEN2. Monitor for neck mass, hoarseness, or dysphagia.	Rodent studies show C-cell tumors. No human evidence; EMA data show no link to other thyroid cancers ^(22,23) .	Contraindicated in MTC/MEN2. Educate on symptoms. Routine calcitonin testing not required.
Hypoglycemia (with insulin/secretagogues)	Monitor glucose in patients on insulin or sulfonylureas.	ESSENCE trial: 7.40% vs 5.40% placebo. Risk higher post-bariatric or with T2D ⁽³⁾ .	Adjust insulin/sulfonylurea dose. Educate on hypoglycemia recognition and management.
Heart rate increase (chronotropic effect)	Check pulse periodically. Ask about palpitations or tachycardia.	GLP-1R activation in SA node increases HR ⁽²⁴⁾ .	Caution in arrhythmia-prone patients. Reassess if persistent tachycardia.
Gastroparesis/delayed gastric emptying	Evaluate baseline motility in at-risk patients. Monitor for early satiety, vomiting, or retained food.	Retrospective incidence: 6.50/1000 person-years ⁽²⁵⁾ . Contraindicated in severe gastroparesis.	Avoid in severe gastroparesis. Consider gastric emptying evaluation if symptoms persist.
Procedural risks (aspiration/retained gastric contents)	Document GLP1-RA use pre-procedure. Coordinate with local anesthesia for fasting interval.	No increased risk of aspiration or respiratory failure found ^(26,29) .	Customize peri-procedural fasting per local guidance. Communicate with anesthesia/endoscopy teams.
Diabetic retinopathy and ocular complications	Baseline if not recently done and periodic ophthalmology evaluation in those with T2DM. Monitor for blurred vision or sudden changes.	SUSTAIN-6 retinopathy 3.00% vs 1.80% placebo. Increased NAION and wet AMD risk ^(30,32) .	Avoid rapid glucose lowering. Refer for any new visual symptoms.
Lean mass loss/sarcopenia	Assess muscle strength/mass (DEXA, grip test). Ensure protein intake and resistance training.	13% lean mass loss (~39% of total weight reduction). 9-10% appendicular loss over 2 years ^(3,33,34) .	Maintain protein 1.20-1.50 g/kg/day. Encourage resistance training. Monitor closely in older/sarcopenic adults.

Abbreviations: AKI, acute kidney injury; AMD, age-related macular degeneration; DEXA, dual-energy x-ray absorptiometry; eGFR, estimated glomerular filtration rate; GI, gastrointestinal; MEN2, multiple endocrine neoplasia type 2; MTC, medullary thyroid carcinoma; NAION, nonarteritic anterior ischemic optic neuropathy; PI, prescribing information; T2DM, type 2 diabetes mellitus.

although there are no data to support an increased aspiration risk in patients on GLP-1 RA therapy. The potential for incomplete endoscopic visualization secondary to retained gastric contents also bears consideration. A large retrospective analysis found that GLP-1 RA use was not associated with increased risk of aspiration pneumonia, respiratory failure, mechanical ventilation, or hospitalization following screening colonoscopy in adults with obesity.^[26] Current evidence is inadequate to establish whether extended fasting or peri-procedural drug interruption mitigates these risks. Nevertheless, documentation of GLP-1 RA treatment at pre-procedure evaluation allows coordination of fasting intervals and anesthesia planning to enhance procedural safety, and clinicians should customize recommendations based on local institutional guidance.^[27–29]

Semaglutide may increase diabetes-related retinopathy complications, particularly in patients with pre-existing diabetic retinopathy. The SUSTAIN-6 trial showed significantly higher diabetic retinopathy complications in semaglutide-treated patients (3.0% vs. 1.8% placebo). Patients with diabetic retinopathy history require careful monitoring, especially during rapid glycemic improvement periods.^[30] A matched cohort study of 16,827 patients found that semaglutide was associated with a 4.28-fold increased risk of nonarteritic anterior ischemic optic neuropathy (NAION) in T2DM patients and a 7.64-fold increased risk in patients with obesity.^[31] A population-based cohort study of patients with diabetes found that those prescribed GLP-1 RAs for 6 months or longer had approximately twice the risk of developing neovascular age-related macular degeneration (wet AMD) compared with unexposed patients with diabetes. While the absolute risk remains low, the risk increases with longer use and in older adults.^[32]

In a subgroup analysis of 140 subjects who underwent dual-energy x-ray absorptiometry (DEXA) scans, those receiving semaglutide lost 6.92 kg of lean mass compared with 1.48 kg in the placebo arm, representing an ~13% reduction in lean mass relative to baseline measures in the total semaglutide population.^[3] Overall, semaglutide therapy results in lean mass loss that constitutes about 39% of total weight reduction, with appendicular skeletal muscle volume declining by 9–10% over 2 years, effects that are particularly pronounced in older adults and those with baseline sarcopenia.^[33,34] Because sarcopenia is associated with poor outcomes in chronic liver disease, ongoing surveillance for muscle loss through physical examination, resistance exercise, and adequate protein

intake (1.2–1.5 g/kg/day) is recommended to mitigate these risks.^[33,34]

Concomitant therapy

Regardless of semaglutide treatment status, best practices in MASLD management should include comprehensive lifestyle modification (nutrition, exercise, and behavior modification) and optimal control of comorbid metabolic conditions.^[1,35,36]

There is minimal information about the combined use of semaglutide and resmetirom. Resmetirom received accelerated approval from the FDA in March 2024 for the treatment of MASH with moderate-to-advanced liver fibrosis consistent with stages F2–F3 fibrosis.^[9,37] The MAESTRO-NASH trial excluded individuals who initiated or dose-modified drugs with GLP-1 RA activity within 6 months of randomization, and 14% of study participants were taking one of these medications at the maximum dose of 1 mg/week for more than 6 months at the time of baseline inclusion biopsy.^[38] GLP-1 RA treated patients showed similar rates of MASH resolution and fibrosis improvement in combination with resmetirom as patients not on these therapies and no notable differences in safety and tolerability were observed.^[39] However, semaglutide and resmetirom have not been studied as a combination therapy for MASH, nor have they been compared for relative efficacy. Exclusion criteria for participation in the ESSENCE clinical trial included treatment with vitamin E (at doses = 800 IU/day) or pioglitazone, which had not been at a stable dose in the opinion of the investigator in the period from 90 days before the baseline liver biopsy until the screening visit. In addition, treatment with weight loss medications that were not taken in stable doses within 28 days of liver biopsy was not permitted.

Assessing response to treatment

Data from the ESSENCE trial are limited regarding changes in specific NITs associated with histological response (Table 1).^[3] As detailed in the AASLD MASLD guidance update for resmetirom,^[9] some data are available to guide clinically meaningful response to therapeutic intervention based upon NITs (though little is known about how changes in NITs correspond to changes in liver histology in clinical trials for treatment

of MASH). Recently published data^[40] reviewed MRI-derived NITs, iron-corrected T1 mapping (cT1), and liver fat content from proton density fat fraction (PDFF) to predict histological improvement in 150 patients who had participated in 1 of 3 clinical trials for MASH. Data from NASH CRN suggest that a change in VCTE-derived LSM over time is independently and bi-directionally associated with risk of liver-related events and, thus, may be used as a surrogate for clinical outcomes in patients with MASLD.^[41] Similar findings have been reported using the Agile 3+ score, which combines VCTE LSM with ALT, AST, platelet count, age, sex, and diabetes status.^[42] Supplemental Table S3, <http://links.lww.com/HEP/K320>, summarizes candidate surrogates for assessing histological response to MASH pharmacotherapy.

Publicly available data from the ESSENCE trial do not allow an assessment of blood tests and other NITs that predict response to treatment (and this is generally an unmet need in clinical trials of MASH pharmacotherapy). Several clinically relevant blood tests (eg, ALT, AST, GGT, HbA1c, lipids) were significantly reduced among patients in the semaglutide arm compared with placebo. The most robust decreases in these parameters occurred during the initial 36 weeks of treatment. Other NIT results from baseline and week 72 are reported according to treatment arm and histological endpoint (summarized in Table 1). Review of these aggregate data suggests that a decrease in ALT and controlled attenuation parameter (CAP) from baseline to week 72 was greater in those treated with semaglutide who achieved MASH resolution without worsening of fibrosis compared with those who did not achieve this endpoint. Relative changes in LSM and ELF score from baseline to week 72 were greater for semaglutide compared with placebo. Based on a collective assessment of available data, a reduction from baseline to 72 weeks of VCTE LSM $\geq 30\%$, MRE LSM $\geq 20\%$, ALT ≥ 17 U/L or $\geq 20\%$, likely reflects a beneficial response. Response below these thresholds but without evidence of worsening or progression to cirrhosis requires individual patient evaluation (Figure 2).

Assessing futility (non-response)

In the ESSENCE trial, 37.1% of participants treated with semaglutide did not resolve MASH, and 63.2% did not have a reduction in fibrosis without worsening of steatohepatitis by week 72. At the current time, we do not have data from the ESSENCE trial to reliably identify patients who are unlikely to resolve MASH or regress liver fibrosis with 72 weeks of semaglutide treatment. In the absence of data, we recommend consideration of previously recommended criteria,^[9]

including lack of decline in ALT or liver stiffness on VCTE, or worsening NIT parameters (Figure 2).

SEMAGLUTIDE AND CARDIOMETABOLIC RISK FACTORS

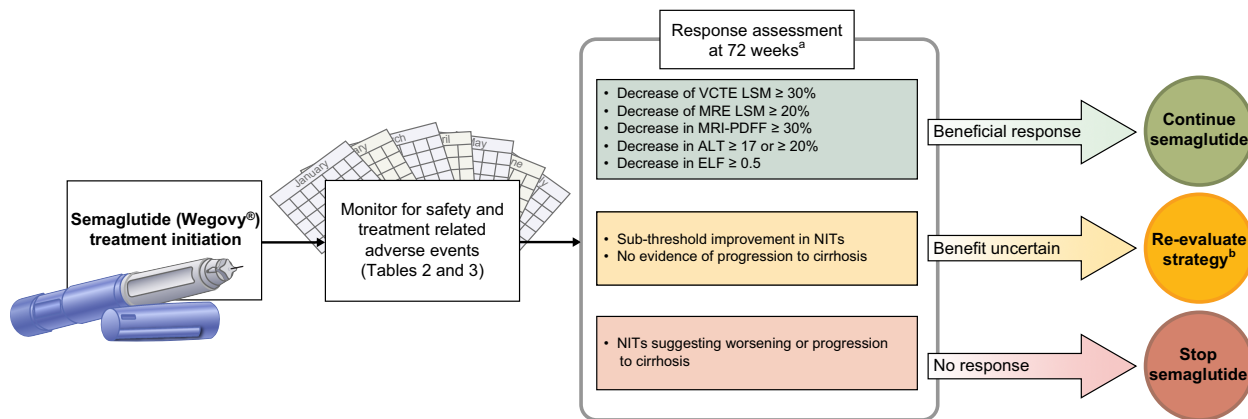
Patients with MASH have a higher risk not only of liver-related morbidity and mortality, but also of adverse cardiometabolic outcomes. Indeed, cardiovascular disease (CVD) is the most common cause of death in patients with MASH.^[43] While the causal relationship between MASH and cardiovascular disease continues to be studied, clinical guidelines have established cardiovascular risk reduction as one of the pillars of comprehensive MASLD management.^[1,17] Management of CVD has largely focused on management of dyslipidemia, hypertension, diabetes (T2DM), and obesity, with the latter two being indications for semaglutide usage in patients with MASLD that predated FDA approval of semaglutide for MASH. While it is likely that semaglutide for MASH would confer similar CV risk-reduction benefits observed in other populations, this remains presumptive as no dedicated outcome trials have yet demonstrated CV benefit specifically in patients receiving semaglutide for MASH.

Semaglutide and obesity

Current MASLD guidance recommends a minimum of 5% weight loss to reduce steatosis, 7%–10% to target inflammation, and at least 10% to reverse fibrosis^[44,45] In addition to diet and exercise, surgical and endoscopic strategies as well as pharmacotherapy with incretin agonists [which include GLP-1 RA and dual glucose-dependent insulinotropic polypeptide (GIP)/GLP-1 RA] are approaches that can be used to address obesity.^[7] The key research on semaglutide 2.4 mg for obesity is encapsulated in the Semaglutide Treatment Effect in People with Obesity (STEP) trials^[46–49] (Supplemental Table S4, <http://links.lww.com/HEP/K321>). These phase 3 global RCTs, which included patients with and without T2DM, form the foundation for both the initiation and maintenance of semaglutide in these individuals, regardless of a MASLD diagnosis. Although these studies were not specifically designed to investigate the benefits of weight loss in a MASLD population, MASLD accounted for 5.7%–21% of the comorbidities among the participants.

The STEP trials (STEP 1–4) collectively demonstrate that semaglutide 2.4 mg weekly is highly effective for weight loss in adults with obesity or overweight, including those with MASLD. Across these trials, semaglutide consistently led to significant reductions

Assessment of safety and response to semaglutide (Wegovy®) for the treatment of MASH with moderate-to-advanced fibrosis (F2-F3)



^a Assess based on the same imaging-based or blood-based markers used to determine treatment eligibility.

^b Options may include re-optimizing lifestyle interventions and considering other therapy, with or without stopping semaglutide.

FIGURE 2 Assessment for treatment outcomes in patients receiving semaglutide. Abbreviations: ALT, alanine aminotransferase; ELF, enhanced liver fibrosis testing; LSM, liver stiffness measurement; MRE, magnetic resonance elastography; MRI-PDFF, magnetic resonance imaging–proton density fat fraction; NIT, non-invasive test; VCTE, vibration-controlled transient elastography.

in body weight, averaging 14.9% in STEP 1, 9.6% in STEP 2 (patients with type 2 diabetes), and 16% in STEP 3 (with intensive behavioral therapy). Importantly, a substantial proportion [from 45.6% (STEP 2) to 75.3% (STEP 3)] of participants achieved $\geq 10\%$ weight loss, the threshold associated with fibrosis reversal in MASLD guidance. In addition, both the STEP 4 trial and STEP 1 extension trials demonstrated that discontinuation of semaglutide led to a reversal of the improvements in cardiometabolic endpoints^[50] indicating that ongoing treatment is necessary to sustain these therapeutic benefits.

Semaglutide and cardiovascular risk reduction

In 2020, the Ozempic formulation of semaglutide (0.5 mg or 1.0 mg weekly) was approved to reduce the risk of major adverse cardiovascular events (ie, CV death, nonfatal myocardial infarction, and nonfatal stroke) in patients with T2DM and established CVD (Supplemental Table S5, <http://links.lww.com/HEP/K322>). This approval was based on the Trial to Evaluate Cardiovascular and Other Long-term Outcomes with Semaglutide in Subjects with Type 2 Diabetes (SUSTAIN-6) trial, which showed a 2.3% absolute risk reduction in these events compared with placebo, with most of the benefit obtained by reducing nonfatal stroke.^[30] In 2024, the Wegovy formulation received FDA approval based on the Semaglutide Effects on Cardiovascular Outcomes in People with Overweight or Obesity (SELECT) trial for similar CV risk

reduction in adults with obesity or overweight and established cardiovascular disease.^[51] SELECT enrolled over 17,000 non-diabetic individuals and demonstrated a 1.5% absolute risk reduction in the composite cardiovascular endpoint after nearly 40 months.

Among patients with MASLD, these CV-related benefits have important implications. MASLD is closely linked to metabolic syndrome and elevated CV risk, making semaglutide's dual benefit of weight loss and CV protection particularly valuable for both diabetic and non-diabetic patients with a history of CV disease. That these benefits were seen irrespective of starting weight or total amount of weight lost in the SELECT trial suggests that there may be weight-independent benefits of semaglutide in this population. In aggregate, these findings support the broader use of semaglutide in MASLD patients not only to meet liver-related weight loss targets but also to mitigate cardiovascular complications, which are a leading cause of mortality in this group.

Semaglutide and chronic kidney disease

In 2025, the FDA expanded semaglutide's indications to include the reduction of CV death risk in patients with T2DM and chronic kidney disease (CKD), as well as the mitigation of kidney disease progression. This decision was based on the Evaluate Renal Function with Semaglutide Once Weekly (FLOW) trial, a phase 3 randomized controlled study involving 3533 adults with T2DM and CKD.^[52] Participants receiving 1.0 mg semaglutide weekly experienced a 24% relative risk

reduction in kidney failure, CV death, and other renal events compared with placebo. The trial was stopped early due to meeting its efficacy endpoints.

These findings are relevant for patients with MASLD, which often coexists with metabolic syndrome, diabetes, and increased CV and CKD risk.^[1] Semaglutide's benefits in reducing both cardiovascular and kidney complications suggest it may offer broader protection for MASLD patients, especially those with overlapping comorbidities.

ENDOCRINOLOGIC CONSIDERATIONS

GLP-1 is an incretin hormone secreted by enteroendocrine L-cells in the distal ileum and colon in response to nutrient intake. GLP-1 RAs are long-acting analogues of human GLP-1 that incorporate structural modifications to slow dipeptidyl peptidase-4 (DPP-4) cleavage, prolong receptor activation, and provide sustained glycemic and appetite control. Semaglutide shares 94% sequence homology with human GLP-1. While native GLP-1 has a half-life of only about 1–2 minutes, semaglutide has a half-life of ~5–7 days, allowing for once-weekly dosing (Wegovy Prescribing Information^[2]).

The GLP-1 receptor (GLP-1R) is expressed across pancreatic β -cells, δ -cells, and α -cells, with widespread distribution on β -cells, moderate expression on δ -cells, and limited expression on α -cells. GLP-1R activation leads to glucose-dependent insulin release from β -cells, stimulates somatostatin secretion from δ -cells, and suppresses glucagon release from α -cells via direct and paracrine mechanisms.^[53] The “incretin effect” refers to the phenomenon in which oral glucose triggers a greater insulin response than intravenous glucose. This occurs because incretin hormones are released in response to nutrients in the gut, leading to increased insulin secretion by the pancreas.^[53–55] GLP-1 RAs stimulate insulin secretion primarily when blood glucose is high, which helps minimize the risk of hypoglycemia (Wegovy Prescribing Information^[2]).

GLP-1Rs are minimally present in cardiac and vascular tissues and are nearly absent on enterocytes. In the liver, GLP-1R expression is negligible on hepatocytes, Kupffer cells, stellate cells, sinusoidal endothelial cells, and most resident immune cells. While it has been suggested that infiltrating $\gamma\delta$ T-cell subsets may express GLP-1R, further studies are needed.^[55,56] Consequently, the hepatoprotective benefits of GLP-1 RAs are thought to be mediated indirectly via sustained weight reduction, improved glycemic and lipid profiles, and reduction of hepatic and systemic inflammation, rather than direct effects on resident liver cells.^[53,57,58]

GLP-1 RAs lead to sustained weight loss largely due to CNS-mediated appetite suppression. GLP-1Rs are

expressed widely in the central nervous system, including the hypothalamus, hindbrain, and brainstem, where they affect neuronal pathways that control satiety and reduce food intake. Activation of these receptors leads to reductions in hunger and caloric consumption. Efficient energy homeostasis relies on bidirectional gut–brain crosstalk through appetite-suppressing enteroendocrine satiety hormones, including GLP-1, glucose-dependent insulinotropic polypeptide (GIP), cholecystokinin (CCK), peptide YY (PYY), and secretin, while pancreatic polypeptide (PP) is secreted by pancreatic islet cells. These satiety signals coordinate with the orexigenic hormone ghrelin, which stimulates hypothalamic agouti-related peptide (AgRP) neurons to drive appetite. Semaglutide modulates gut–brain crosstalk by enhancing glucose-dependent insulin secretion, slowing gastric emptying, and activating hypothalamic satiety centers.^[57–59]

Through improved glycemic control and promoting sustained weight loss, GLP-1 RAs enhance insulin sensitivity in adipose tissue and muscle, accompanied by reductions in pro-inflammatory cytokines (TNF, IL-1, IL-6) and elevations in adiponectin. GLP-1 RAs also have beneficial effects on lipid metabolism and hepatic fat content through multiple mechanisms. They inhibit chylomicron synthesis and secretion, decrease de novo lipogenesis, and reduce hepatic production of very-low-density lipoproteins (VLDL). In addition, the weight loss induced by GLP-1 RAs helps lower circulating levels of free fatty acids and ceramides.^[17,18,57,58]

SUMMARY

FDA-indication and practice recommendation:

- Semaglutide (Wegovy formulation) received accelerated FDA approval in August 2025 for treating MASH with moderate-to-advanced fibrosis (consistent with stages F2–F3 fibrosis), based on phase 3 ESSENCE interim trial results where 72 weeks of 2.4 mg/week subcutaneous injection resulted in achievement of both primary histologic endpoints: (1) resolution of MASH without worsening of fibrosis (62.9% vs. 34.3% placebo, $p < 0.001$) and (2) ≥ 1 stage reduction in liver fibrosis without worsening of MASH (36.8% vs. 22.4% placebo, $p < 0.001$); final approval awaits long-term outcomes.
- This AASLD update provides practical recommendations for identifying, selecting, and monitoring patients with MASH eligible for semaglutide (Wegovy), integrating data from ESSENCE and prior clinical studies to complement FDA labeling.

Patient selection:

- Semaglutide (Wegovy) candidates should have MASH with stages 2–3 fibrosis, identified using

non-invasive tests (NITs) such as VCTE (8–15 kPa), MRE (3.1–4.4 kPa), or ELF (9.2–10.5), rather than liver biopsy, which is impractical and unnecessary for most patients. In those with VCTE (15–20 kPa), MRE (4.4–5 kPa), or ELF (10.5–11.3), an individualized decision to treat should be based on exclusion of cirrhosis with another confirmatory NIT, or cross-sectional imaging excluding nodular-appearing liver contour and signs of portal hypertension, or a platelet count of $<150,000/\text{mm}^3$.

- Semaglutide is not approved as an indication for patients with MASH cirrhosis (VCTE >20 kPa, MRE >5.0 kPa, ELF >11.3 , and/or evidence of portal hypertension). Those with compensated cirrhosis on semaglutide for another FDA-approved indication should be monitored carefully.

Monitoring and safety:

- Semaglutide showed a favorable hepatic safety profile in the ESSENCE trial, with no discontinuations due to liver enzyme elevations. Routine hepatic panels are recommended only as clinically indicated. The most common adverse events were GI (nausea, diarrhea, constipation, vomiting), generally mild and transient; patient education and dose titration help improve tolerance. Clinicians should monitor for rare but serious risks, including acute kidney injury (from dehydration), symptomatic gallbladder disease, pancreatitis, thyroid C-cell tumors, retinopathy progression, and lean mass loss.

Treatment response and concomitant therapy:

- Lifestyle modification remains the cornerstone of MASLD/MASH management alongside semaglutide. Combination use with resmetirom at the dose of 2.4 mg/week has not been studied. While no NITs are known to reliably predict histologic response at the individual patient level, reductions from baseline to 72 weeks of treatment suggest significant improvement in MASH resolution (ALT ≥ 17 U/l or $\geq 20\%$) and fibrosis improvement (VCTE LSM $\geq 30\%$; MRE LSM $\geq 20\%$; ELF ≥ 0.5). Non-response may be suspected if ALT or NITs worsen. The benefit is uncertain if the suboptimal response.

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AUTHOR CONTRIBUTIONS

All authors contributed equally to this work.

CONFLICTS OF INTEREST

Rotonya M. Carr consults for Novo Nordisk, Intercept, and Madrigal. The remaining authors have no conflicts to report.

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